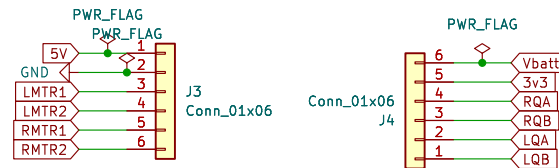
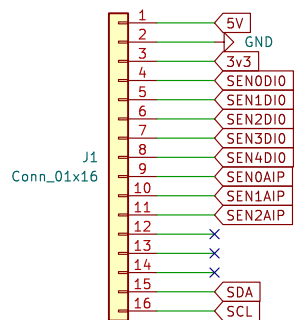


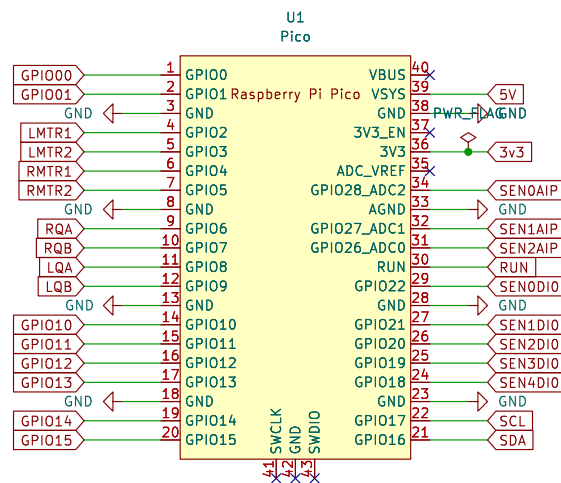
Sensor connector



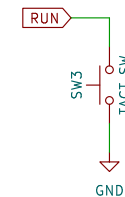
Mezzanine connections

MicroController voltage is 3.3v for any RP2040

Raspberry Pi Pico / PicoW

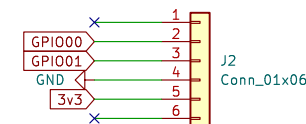


PiPico RESET push button

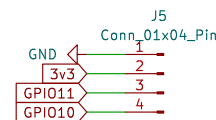


Breakout Connectors

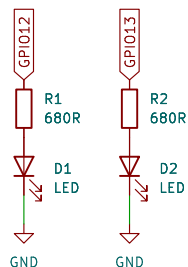
J2 presents GP0 and GP1 with 3v3 and Gnd on pins suitable for bluetooth module to communicate on UART0. The pins can also present I2C0
A vs838 IR receiver could be installed on J2.



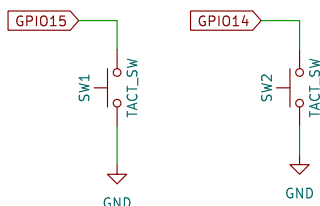
J5 presents GP10 and GP11 for I/O or alternative connection to I2C1 suitable for connecting 0.9" OLED screen or KY-022 IR receiver



Board LEDs



Board switches/push buttons



MECHANICAL



UKMARS Gemini RP2040 Mezzanine Board Schematic

Sheet: /

File: UKMARSBOT_RP2040Mezzanine.kicad_sch

Title: UKMARS Gemini RP2040 Mezzanine V2.0

Size: A4

Date: 2025-07-07

Rev: 2

KiCad E.D.A. 9.0.7

Id: 1/1